MIT MYSORE

System Architecture and Workflow

Maharaja Institute of Technology Mysore Examination System

PROJECT SCOPE

Prepared as the shared reference for the MIT Mysore examination-system frontend and backend teams. The official college logo will be inserted in the top-left brand position after it is supplied.

WORKING DESIGN DOCUMENT

Version 1.0 | 12 November 2026

1. System purpose and scope

The Maharaja Institute of Technology Mysore Examination System is an examination-management platform for institutional administration, question-bank preparation, examination scheduling, secure offline delivery, MCQ evaluation and controlled result release. The team nickname is not a product name and must not appear in the system UI.

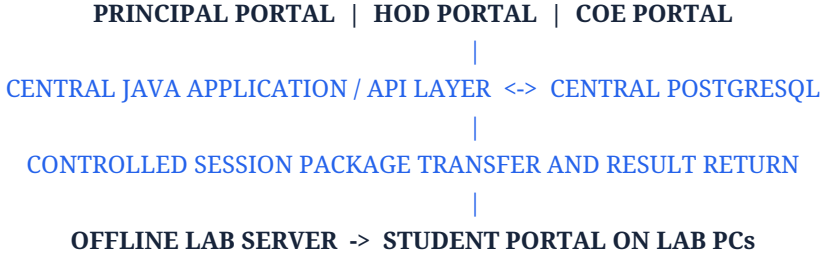
The design separates the central administrative environment from the college laboratory environment. Principal, HOD and COE users work through role-specific portals. Students use a local examination portal connected to a lab server, so the examination can continue without internet access.

> **ARCHITECTURE STATUS**
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> This document is a working architecture blueprint. It describes the agreed workflow and the recommended separation of services. Final API contracts, deployment topology, cryptographic implementation and lab transfer method must be confirmed by the development team.

2. Main system components

Component	Primary responsibility
Principal / Central Admin Portal	Set up the college structure, departments, semesters, sections, subjects, accounts and institution-level access.
HOD / Department Admin Portal	Upload the subject question bank, answer keys and student details; validate and finalize the department submission.
COE Portal	Create examination cycles, schedules and sessions; select prepared papers; manage secure packages, lab transfer, result import and result approval.
Central Java application services	Expose APIs, enforce roles, validate inputs, manage records and coordinate paper generation and result administration.
Central PostgreSQL database	Store institutional data, question-bank revisions, prepared papers, rosters, examination sessions, package metadata, results and audit records.
Lab server	Verify and hold the encrypted session package, activate the session at the scheduled time, distribute question versions locally, collect answers and evaluate MCQs.
Student portal	Authenticate against the lab roster, display instructions and assigned questions, capture answers and submit responses.

3. High-level architecture



4. Institution setup: Principal / Central Admin Portal

1. The Principal or authorized central administrator signs in to the administrative portal.
2. The administrator creates or maintains the academic year, departments, semesters, sections and subjects.
3. The administrator creates accounts for HODs and assigns each HOD to the correct department.
4. The administrator configures authorized COE accounts and institutional permissions.
5. The system stores the structure and access assignments in the central database.

> **ACCESS RULE**
> A CSE HOD must only manage the CSE records assigned to that HOD. Department boundaries must be enforced by the backend, not only hidden in the frontend.

5. Question-bank and student preparation: HOD Portal

1. The HOD signs in and selects the assigned department, semester, section and subject.
2. The HOD uploads 60 questions for each of the five modules, for a total of 300 questions per subject question bank.
3. Each MCQ includes question text, four options, the correct answer, module number and a unique Question ID. The HOD is expected to submit questions of comparable difficulty.
4. The backend validates required fields, question counts, option completeness, answer presence, module mapping and duplicate or suspicious records. The HOD reviews any validation errors.
5. The HOD uploads the student roster for the relevant department, semester and section. The agreed roster fields are USN, student name and date of birth.
6. The HOD reviews the upload summary and performs final submission.
7. The backend records a finalized question-bank revision and prevents ordinary HOD editing of that revision. Any correction must follow an authorized revision or reopening process.

6. Recommended paper-generation stage

The working design is a two-stage generation process. First, after the HOD finalizes the question bank, the backend Question Paper Generation Engine prepares four numbered paper sets. Later, when the COE selects one paper for a session, the Version Generation Engine creates four lettered versions of that selected paper.

Stage	Operation	Output
Stage A	After HOD finalization, select six questions from each of five modules for every numbered paper. Prefer no overlap between the four papers when enough questions exist.	Papers 1-4; 30 questions each.
Stage B	When COE selects a numbered paper, generate four versions with the same 30 questions in different question order. Option order may also be shuffled.	Versions A-D for that selected paper.

7. COE examination-cycle and scheduling workflow

1. The COE signs in and creates an examination cycle, such as an Internal Assessment or semester examination.
2. The COE selects participating departments, semesters, sections and subjects.

3. The system supports a common schedule for participating groups or staggered schedules where departments or sections have different timings.
4. For each subject sitting, the COE sets the date, start time, duration, morning or afternoon session and assigned laboratory.
5. The backend checks scheduling conflicts and verifies that the required question-bank revision and student roster are finalized.
6. When the session is finalized, the backend freezes a session-specific snapshot of the eligible student roster, question-bank revision, schedule and exam settings. Other departments or sections are not changed by this freeze.

8. Paper selection and A-D version creation

The COE selects a prepared numbered paper for each examination session. The working example is Paper 1 for the morning and Paper 2 for the afternoon. The two sessions are independent and each receives its own session identity, roster snapshot, package and result record.

HOD FINALIZED QUESTION BANK

BACKEND PREPARES PAPER 1 / 2 / 3 / 4

COE SELECTS A NUMBERED PAPER FOR A SESSION

VERSION ENGINE CREATES A / B / C / D

FOUR VERSIONS ARE STORED WITH QUESTION-ID MAPPING

The COE can see paper metadata, version-generation status and session association, but cannot read protected question text or correct answers. Every question and answer key remains linked by a stable Question ID, not by its displayed position, so shuffling does not break evaluation. The backend enforces this restriction.

9. Secure session package creation

For each examination session, the backend creates one secure package containing the four completed versions A-D, answer keys mapped by Question ID, the frozen student list for that session, the exam settings and the unique examination/session identifiers.

- Encrypt the package for the intended lab server so the COE cannot read its protected contents.
- Digitally sign the package using an authorized signing key held by the central backend.
- Include identifiers, revision numbers and integrity metadata so the lab can reject the wrong or outdated package.
- Keep package creation and transfer status distinct: prepared, exported, received, signature verified and accepted are different states.

> **CRYPTOGRAPHY BOUNDARY**
>
> Recommended pattern: encrypt to the lab server public key and sign with the central service private key. The lab decrypts with its protected private key and verifies using the trusted central public key. Key storage and rotation belong in trusted server code, never browser JavaScript.

10. COE-to-lab transfer and lab activation

1. The COE exports the encrypted and signed session package from the COE portal.
2. The package is transferred to the designated lab server through the college-approved isolated connection or physical transfer process. The exact method must be selected by the institution.
3. The lab server checks the package signature, identifiers, revision and integrity. An invalid or mismatched package is rejected and logged.

4. A valid package is stored in encrypted form on the lab server until the scheduled start time.
5. At the scheduled time, the lab server checks the session schedule and activation conditions, then decrypts the required examination data internally.
6. The lab server maps the four versions to laboratory computers using a configured computer-number assignment. The mapping is recorded for later evaluation.
7. Student computers receive only the question content and session information required to conduct the examination. The answer key is not sent to student browsers.

11. Offline student examination workflow

1. A student opens the local Student Portal from an assigned laboratory computer.
2. The student enters USN and date of birth. The local lab server checks the values against the frozen roster in the session package.
3. The server confirms that the student is eligible, the session is active and the student has not already completed the examination.
4. The student reads the instructions and starts the exam.
5. The lab server provides the version assigned to that computer. The student cannot select or change the version.
6. The student answers MCQs using the question navigation interface. Responses are saved locally to the lab server against stable Question IDs.
7. The student submits manually, or the lab server closes and submits the attempt when the authoritative examination timer expires.
8. The student sees a completion confirmation, not marks or the answer key.

> **OFFLINE OPERATION**
> The exam must not depend on internet access. Student browsers communicate with the lab server over the local laboratory network. The lab server, not the browser clock, controls the authoritative exam start and end time.

12. Answer collection and evaluation

1. The lab server collects submitted responses from the participating computers.
2. Each response is associated with the student, session, assigned version and Question ID.
3. The lab server evaluates MCQ responses against the protected answer key using the configured marking scheme.
4. The system records correct, incorrect and unanswered counts, marks and submission status.
5. The lab server creates a result file containing the session identity, student results, relevant response records and verification metadata.
6. The result file is digitally signed by the lab server so the COE system can verify its origin and integrity.

13. Result import, COE approval and HOD release

1. The COE retrieves the result file from the lab server using the approved return-transfer method. The same designated connection may be used for sending and retrieving files.
2. The COE portal verifies the lab signature, examination/session identity, roster match and file integrity before import.
3. The COE reviews attendance, absent count, highest marks, average marks, pass percentage and individual results.

4. Results remain under COE control until the COE approves and releases them.
5. After release, each HOD can view only the results for the HOD's own department and authorized sections.
6. The Principal can view institution-level reports according to assigned permissions. Student-facing results remain unavailable under the current requirement.

14. Paper retirement and record retention

Record	After the session
Selected numbered paper	Mark as consumed/retired and remove it from future selection and lab availability.
Master question bank	Retain in the central database with its finalized revision history.
Answer keys	Retain securely in the central record with restricted access.
Session package	Retain package metadata and required audit evidence; remove active lab availability after authorized cleanup.
Student responses and result	Retain according to institutional record-retention rules.
Audit records	Retain creation, transfer, verification, activation, submission, import and approval events.

15. Key data entities

Entity	Purpose / relationships
Department / academic structure	Department, semester, section and subject records.
User / Role	Principal, HOD, COE and authorized staff identities and permissions.
Student	USN, name, date of birth and academic assignment.
Question-bank records	Finalized revision, questions, options and protected answer keys linked by Question ID.
GeneratedPaper / PaperVersion	Numbered paper 1-4 and lettered A-D ordering for a selected paper.
ExaminationCycle / Session	Examination, sitting, schedule, subject, section and status.
Roster / ComputerAssignment	Frozen eligible students and lab computer-to-version mapping.
SecurePackage / Transfer	Package identity, cryptographic metadata and transfer status.
Response / Result	Student responses, evaluation outcome and approval/release state.
AuditEvent	Append-only record of important user and system actions.

16. Frontend and backend integration boundaries

The frontend must communicate with Java services through documented APIs. It must not connect directly to PostgreSQL. All authorization, validation, question generation, answer-key handling, cryptographic operations and result approvals must be enforced on trusted server-side services.

Frontend area	Backend/API responsibility
Principal portal	Institution structure, departments, semesters, subjects, staff and permissions.
HOD portal	Question upload, student roster upload, validation, finalization and upload history.
COE portal	Examination cycles, scheduling, roster snapshots, paper metadata, version generation, package export, result import and approval.
Student portal	Local login, session instructions, assigned version retrieval, response saving and submission.
Lab management	Package verification, activation, computer mapping, local monitoring, answer collection and signed result export.

Suggested API domains

- Authentication and role/permission management.
- Institution structure and academic assignments.
- Question banks, revisions and validation.
- Generated papers and lettered versions.
- Examination cycles, sessions, scheduling and roster snapshots.
- Secure package export and transfer status.
- Lab server activation, student authentication and response capture.
- Result import, verification, approval and reporting.
- Audit and operational status.

17. Security and reliability requirements

- Enforce role-based access on every protected API, not only in navigation or page rendering.
- Keep passwords, signing keys, decryption keys and answer keys out of browser code and frontend bundles.
- Use transport security for central API connections and controlled access to the lab network.
- Use cryptographic signing and encryption with protected key storage on trusted services.
- Use immutable or append-only audit records for critical actions.
- Prevent duplicate student attempts and preserve responses safely during local network interruptions.
- Use authoritative server-side timing for examination activation and closure.
- Back up the central database and define lab-server recovery procedures.
- Use fictional test records during frontend and integration development.

18. Decisions to confirm before implementation is frozen

Decision	Why it must be confirmed
Paper 1-4 generation timing	Working design is generation after HOD finalization. Confirm this matches the backend plan.
Paper overlap policy	Confirm whether questions must be unique across all four numbered papers or only minimized where possible.
Option shuffling	Confirm whether A-D versions shuffle options as well as question order.

Decision	Why it must be confirmed
Lab transfer method	Select USB, isolated Ethernet or another controlled method and define custody procedures.
Lab activation key handling	Define how the lab server stores and uses its decryption key securely.
Offline COE monitoring	Confirm whether COE is physically connected to the lab network during the exam or sees status after synchronization.
Result and audit retention	Confirm institutional retention, backup and deletion rules.
Late/special student access	Define how approved late attempts or exceptional permissions are recorded and restricted.

IMPLEMENTATION PRINCIPLE

Build the portals around stable workflows and API contracts. Keep central administration separate from offline exam runtime. Treat each examination session as an independently identified and frozen unit.

End of System Architecture and Workflow